Assigning Static IPs to NLBs with ALB Target Groups
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Introduction
In this lab, I configured a multi-tier Elastic Load Balancer (ELB) architecture to combine the intelligent routing capabilities of an Application Load Balancer (ALB) with the static IP addressing capabilities of a Network Load Balancer (NLB). This approach allows external DNS systems, which cannot use AWS-specific alias records like those available in Route 53, to reference load balancers through consistent IPs.

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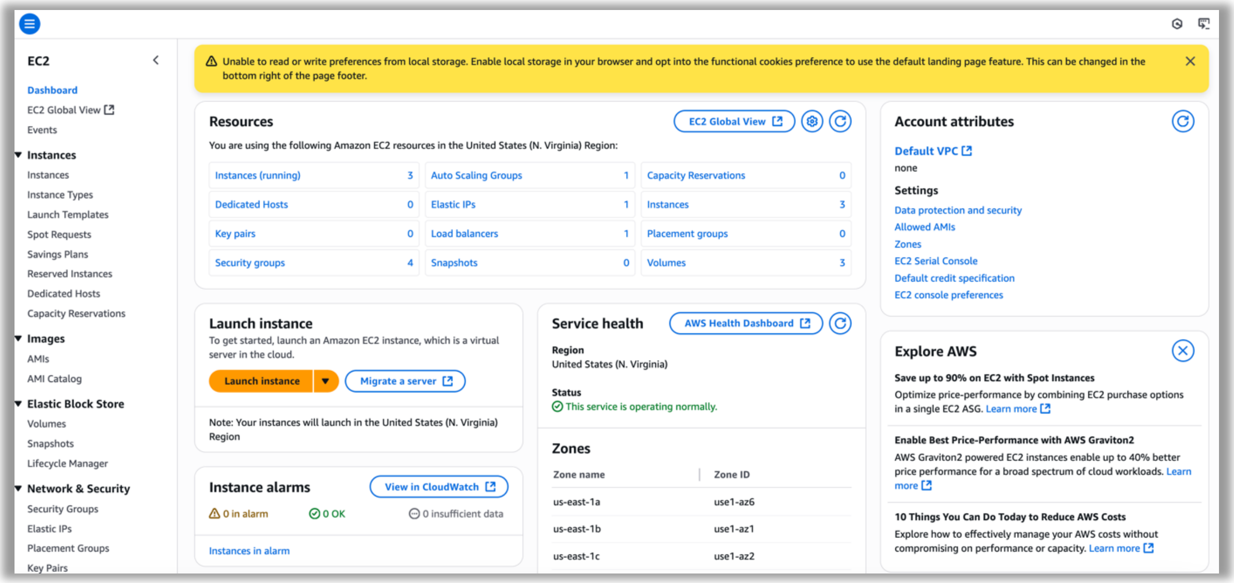
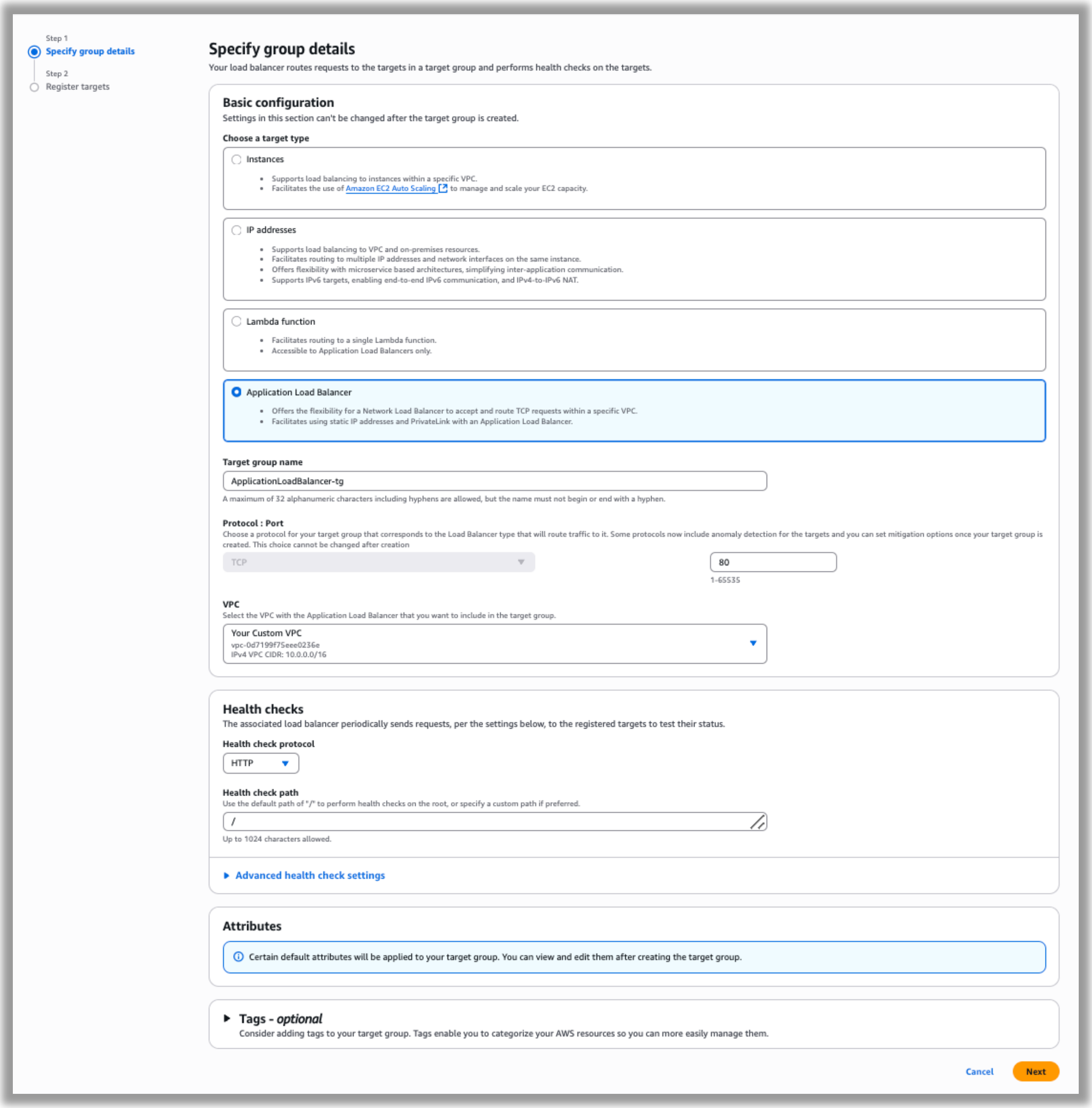
1. Created a Target Group for the ALB



Assuming an existing Auto Scaling Group (ASG) was already running with two EC2 instances behind an ALB, I proceeded to create a target group pointing to this ALB.

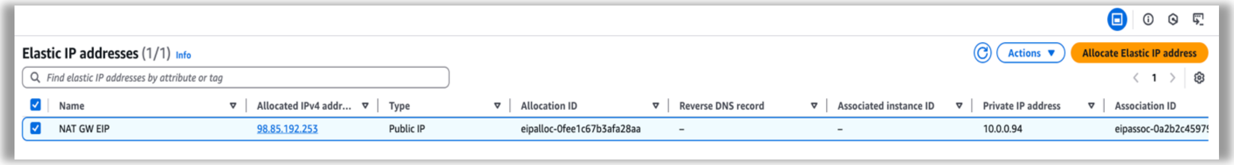


- In the EC2 console, under Load Balancing > Target Groups, I selected Create target group.
- For the target type, I chose Application Load Balancer.
- I named the target group ApplicationLoadBalancer-tg.
- Set the Protocol to TCP and Port to 80.
- Selected the appropriate custom VPC from the dropdown list.
- For health checks, I configured the protocol as HTTP and set the path to /.
- Proceeded to register the target by selecting Register now.
- From the Application Load Balancer dropdown, I selected OurApplicationLoadBalancer.
- Clicked Create target group to complete the process.

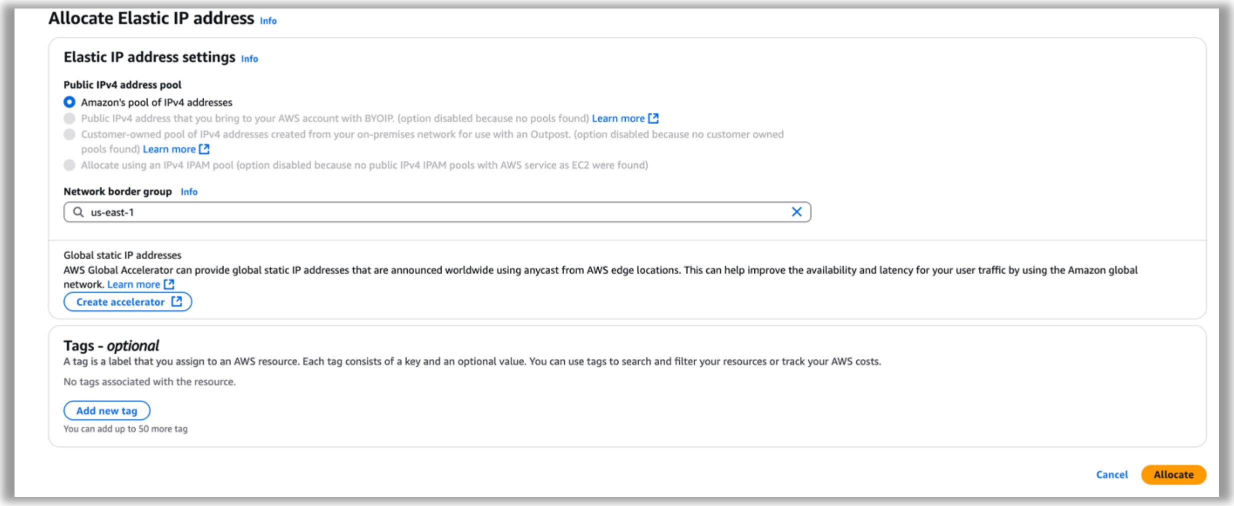
2. Allocated Elastic IP Addresses (EIPs)

To ensure static IP addressing for the NLB, I allocated Elastic IPs:

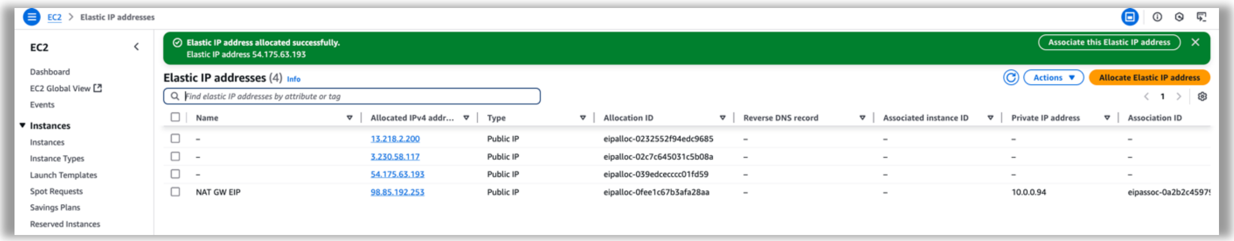
Navigated to Elastic IPs under Network & Security.



Clicked Allocate Elastic IP address.



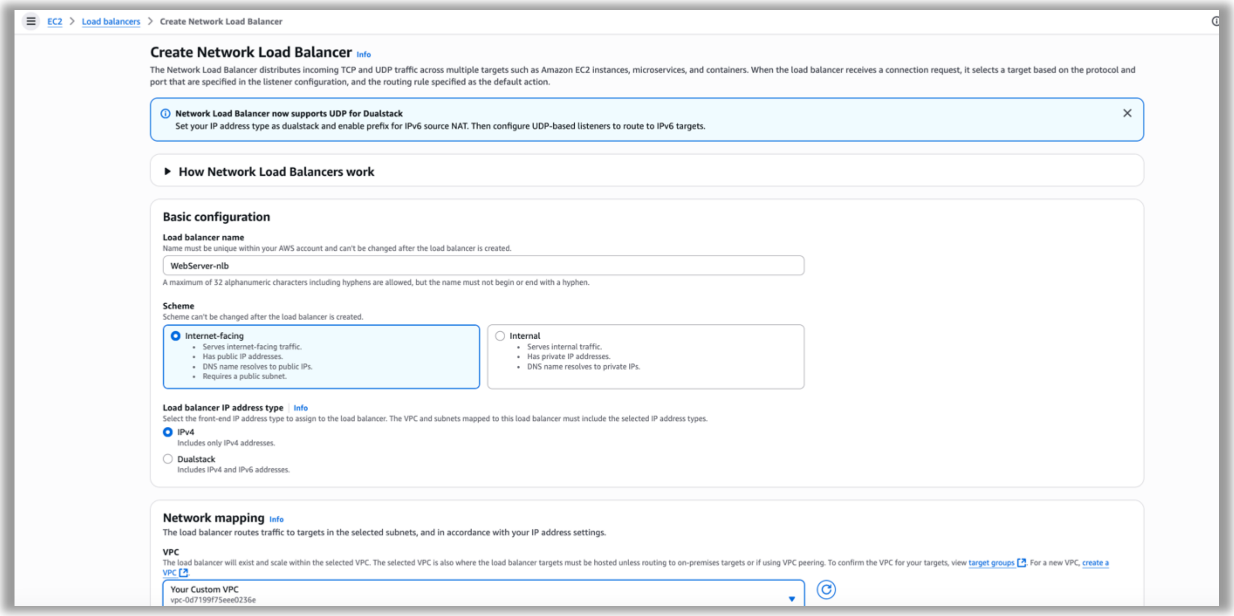
- Ensured the Network Border Group was set to us-east-1.
- Left Amazon's pool of IPv4 addresses selected.
- Repeated the allocation process three times, recording each EIP for later use.



3. Created the Network Load Balancer (NLB) and Listener

With EIPs and a target group ready, I proceeded to configure the NLB:

- In the Load Balancers section of the EC2 console, I clicked Create load balancer and chose Network Load Balancer.
- Named it WebServer-nlb.
- Selected Internet-facing for the scheme.
- Kept IPv4 for IP address type.

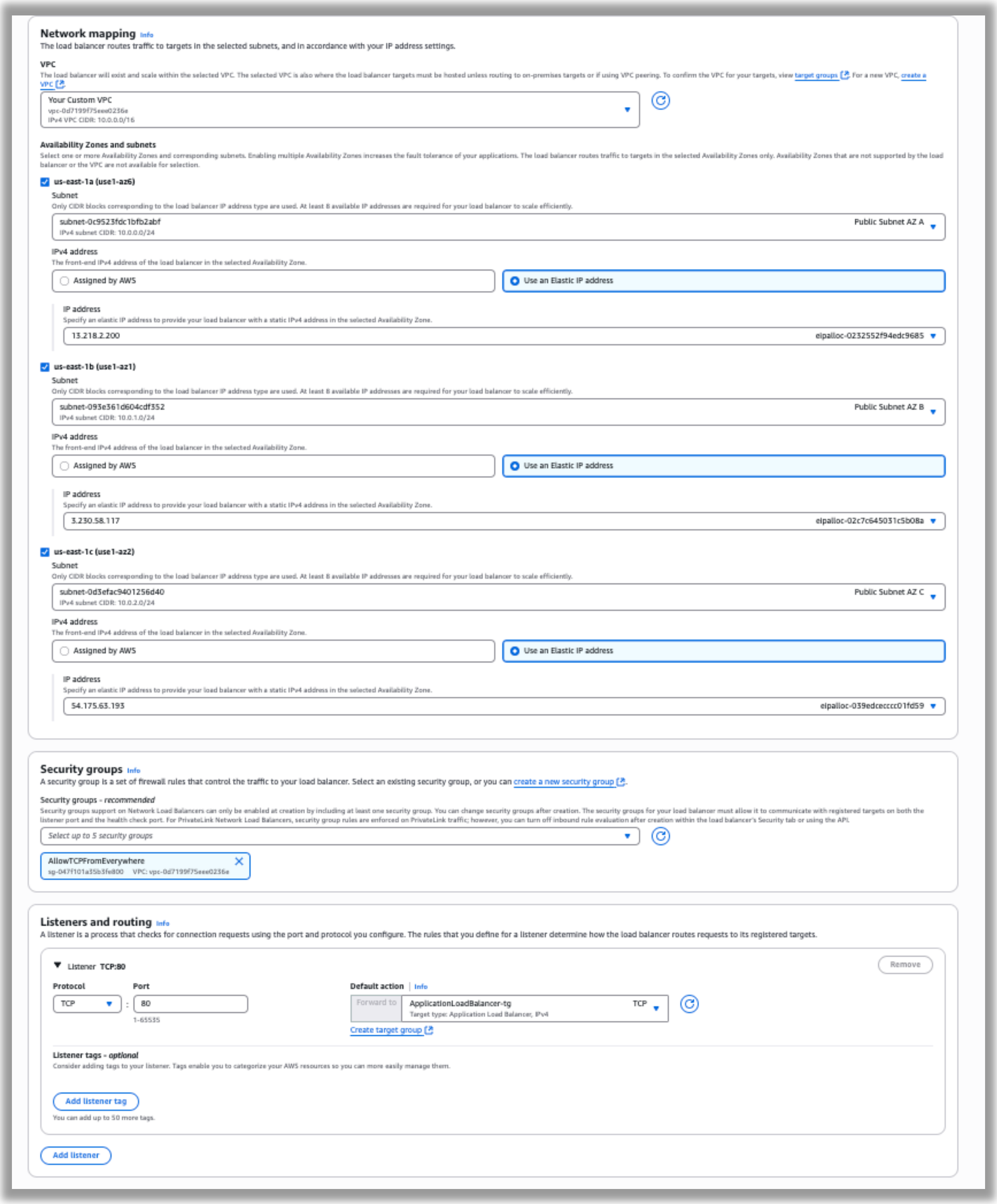


Network Mapping

- Selected the custom VPC (10.0.0.0/16).
- Enabled all three Availability Zones and selected a Public Subnet for each.
- For each subnet, selected Use an Elastic IP address, then assigned a unique EIP from the ones allocated earlier.

Security Groups and Listener

- In the Security groups section, selected the AllowTCPFromEverywhere group and removed the default.
- Under Listeners and Routing, kept the listener configuration at TCP:80.
- For the default action, selected the previously created ApplicationLoadBalancer-tg.
- Clicked Create load balancer and waited for WebServer-nlb to reach the Active state.



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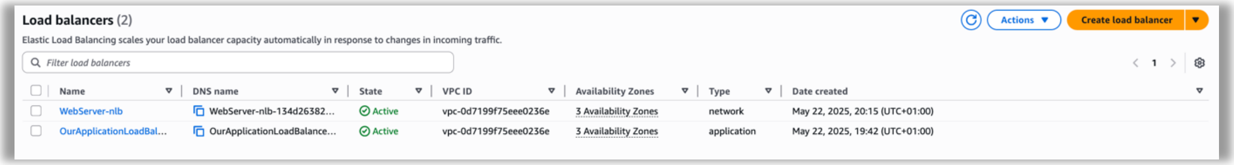
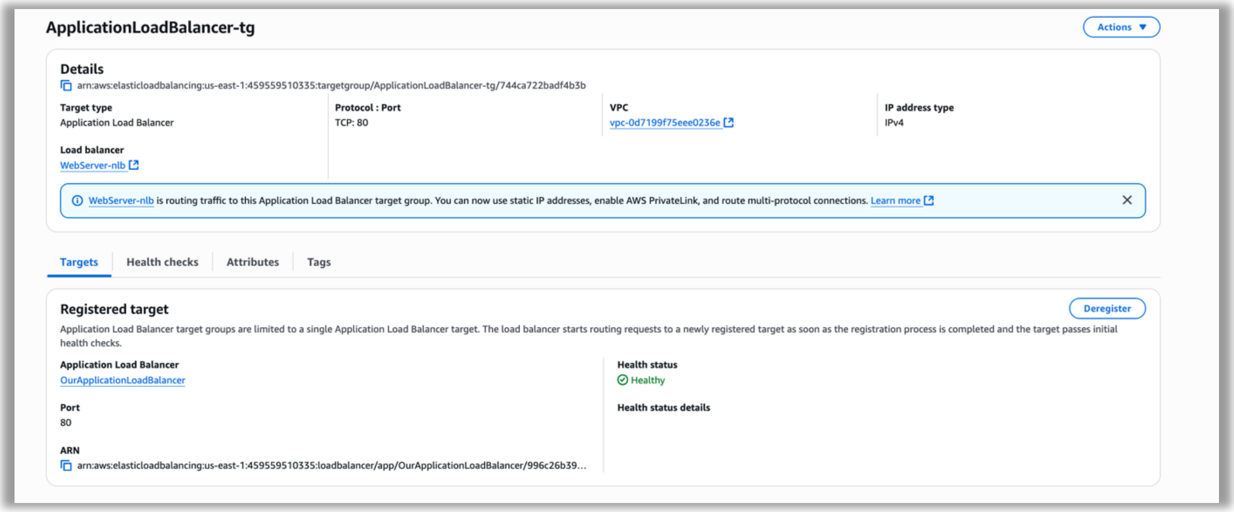
4. Verified Functionality

Once the NLB became active:

- I copied the DNS name of WebServer-nlb and accessed it via a browser to confirm proper functionality.
- Navigated back to the EC2 > Load Balancers section, selected the NLB, and under the Listeners tab, opened the associated Target Group in a new tab.



Verified that the target health status was marked as Healthy, confirming that the NLB was successfully routing traffic to the ALB and onward to the EC2 instances behind it.



Conclusion

This setup successfully leveraged a Network Load Balancer to provide static IP addresses for external DNS systems, while routing traffic to an Application Load Balancer capable of intelligent HTTP routing. The architecture resolved the challenge of integrating with DNS platforms that cannot use AWS ALB alias records, enabling static, reliable endpoint management for web applications hosted on EC2 behind an ALB.